



DRONE CHALLENGE (UAV)

FORT WORTH, TEXAS

TEAM: 2214-1

2025

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Photo Log (C)



**ALL OPEN SOURCE
MATERIAL SHIPPED
AND READY TO BE
ASSEMBLED**



**ASSEMBLING THE
CARBON-FIBER
FRAME OF THE
DRONE**

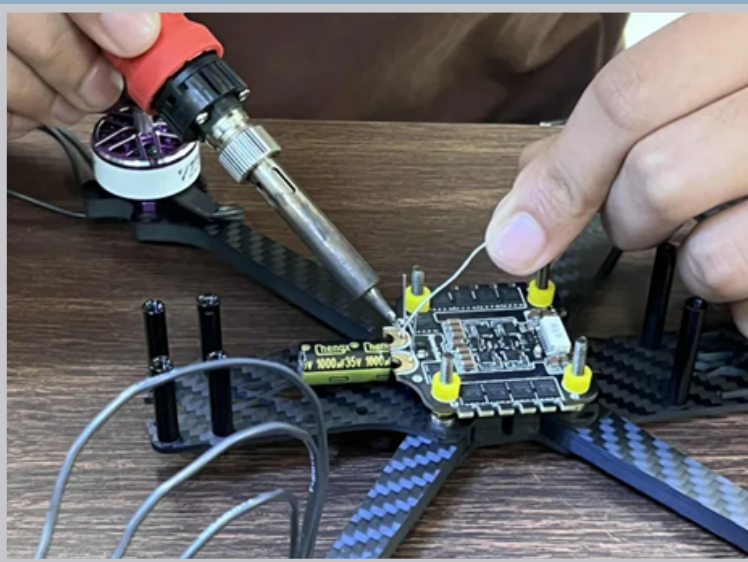


**ATTACHING THE
MOTORS TO THE
DRONE FRAME**

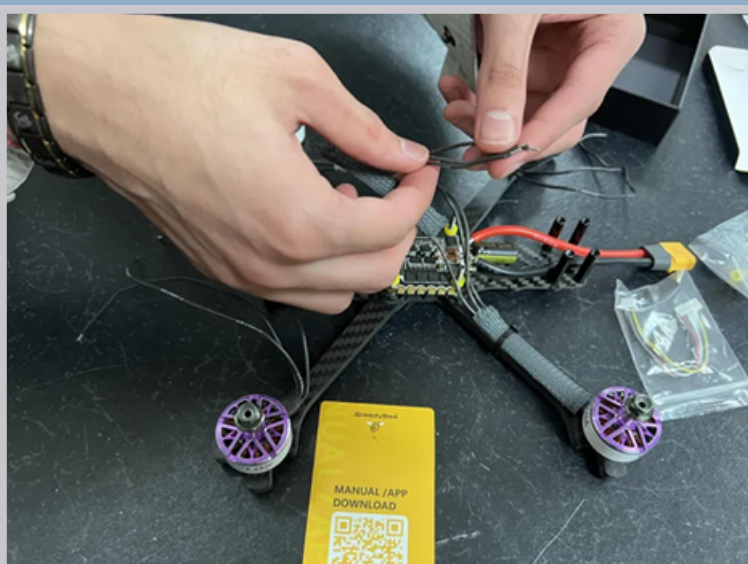
Photo Log (C)



**PUTTING THE
SPEEDYBEE FLIGHT
CONTROLLER
STACK ON THE
FRAME OF THE
DRONE**

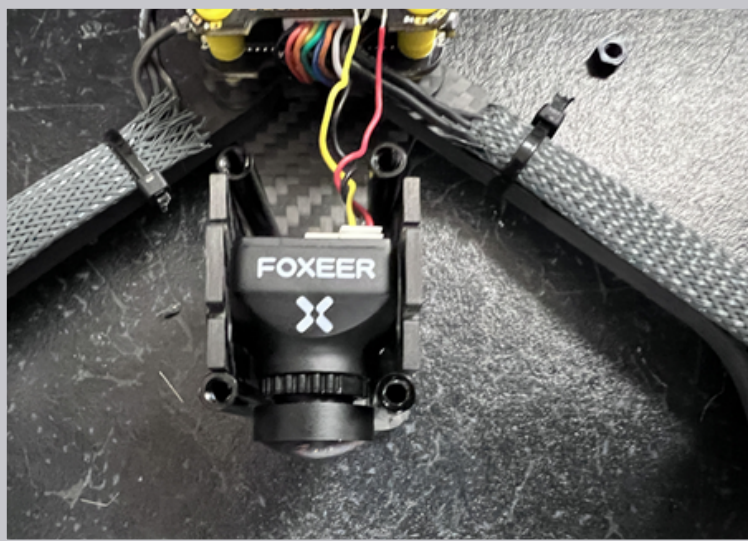


**SOLDERING THE
CAPACITOR ON THE
FLIGHT
CONTROLLER
STACK**

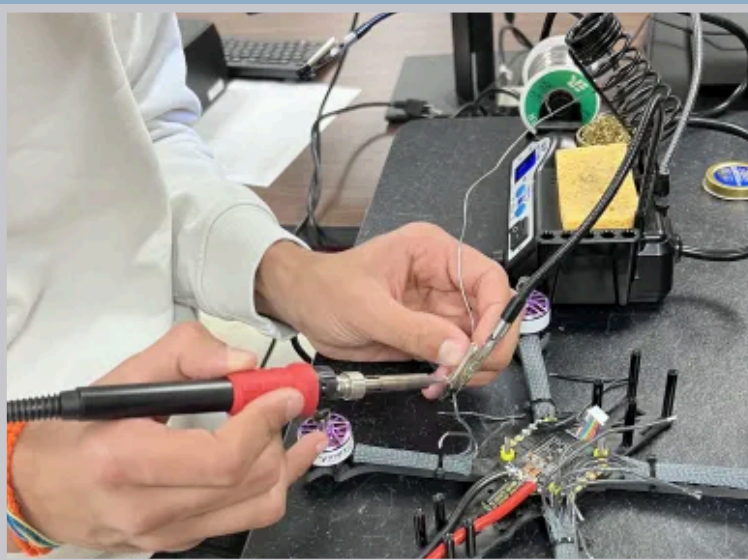


**PUTTING THE WIRES
OF THE MOTORS
INSIDE OF MESH
COVERING**

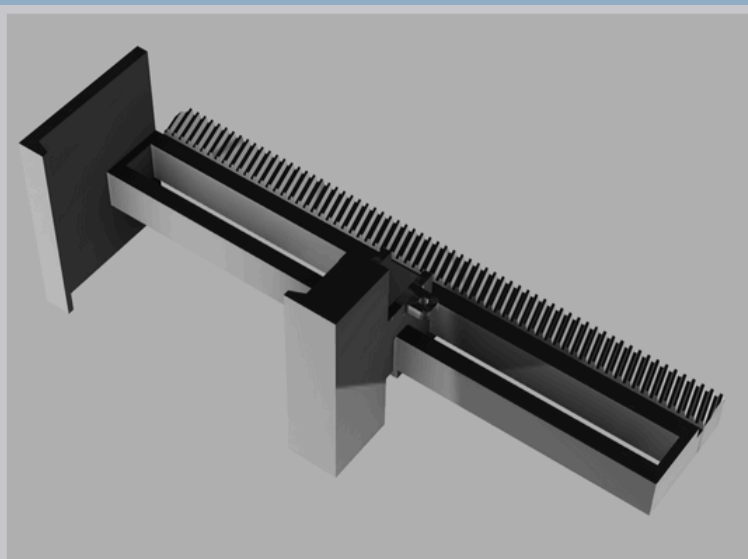
Photo Log (C)



**ATTACHING THE
CAMERA TO THE
FLIGHT
CONTROLLER**

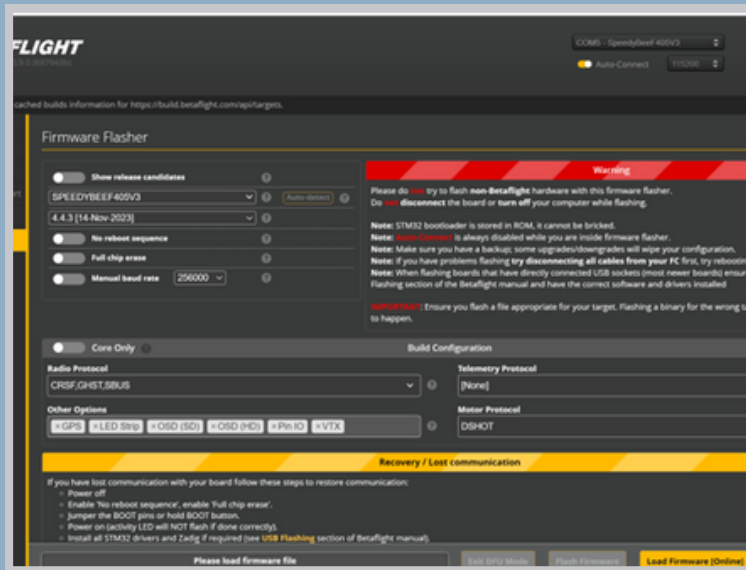


**ATTACHING VIDEO
TRANSMITTER (VTX)
AND RECEIVER (RX)
TO DRONE**

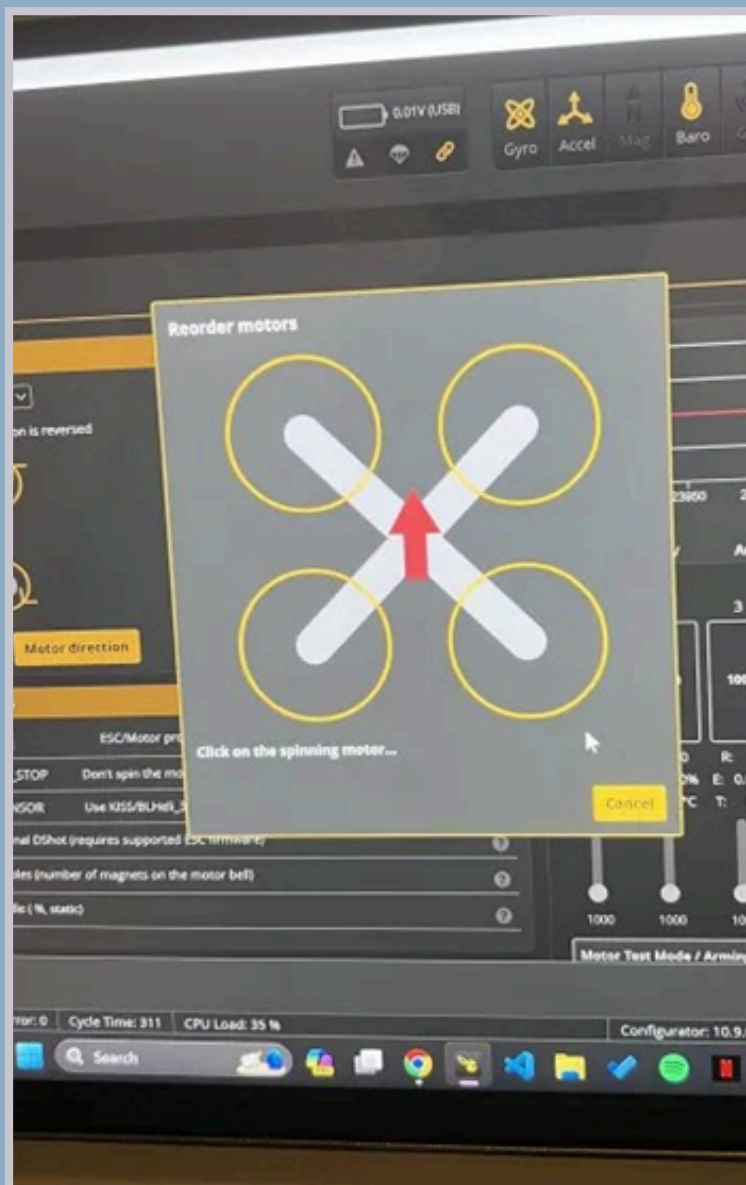


**3-D MODELED
GRABBER CLAW FOR
THE DRONE**

Photo Log (C)



FLASHING CURRENT 3.0.1 EXPRESS LRS SOFTWARE TO DRONE



CONFIGURING DRONE MOTOR DIRECTION AND TESTING SPIN

Photo Log (C)



**PAIRING THE RADIO
TO THE RECEIVER
ON THE DRONE**

1:40

5G 67

Where would I find that, in radio settings or betafight? Sorry I'm a bit new to this. :)

PM Edit Reply

 **SnowLeopardFPV** 777

(44 minutes ago)

If you are getting a solid green LED then the transmitter and receiver are binding. If you had model match enabled and the receiver model number didn't match the one the transmitter was expecting then it wouldn't even bind and you wouldn't even get a solid green LED. Likewise with the binding phrase. If there was a mismatch you wouldn't even get a sold green LED "bound" status.

It's more likely to be a Betaflight settings or a wiring issues between the receiver and the FC.

Can you please run the "dump", "diff all", and "status" commands in the Betaflight Configurator CLI tab and copy/paste the results back here.

Please also post some good clear photos of the actual wiring and connections between your receiver and the flight controller.

PM
Like
Reply

« Next Oldest | Next Newest »

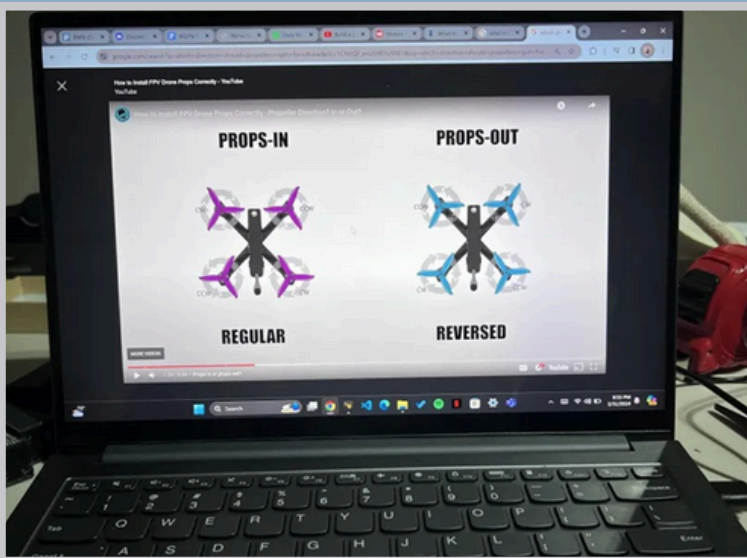
Post Reply

**CONVERSING ON A
TECHNICAL FORUM
TO DEBUG A FAULTY
RECEIVER WHICH
WAS NOT TAKING
INPUTS**

Photo Log (C)



**CONNECTING THE
GOGGLES WITH THE
VTX ON THE DRONE**

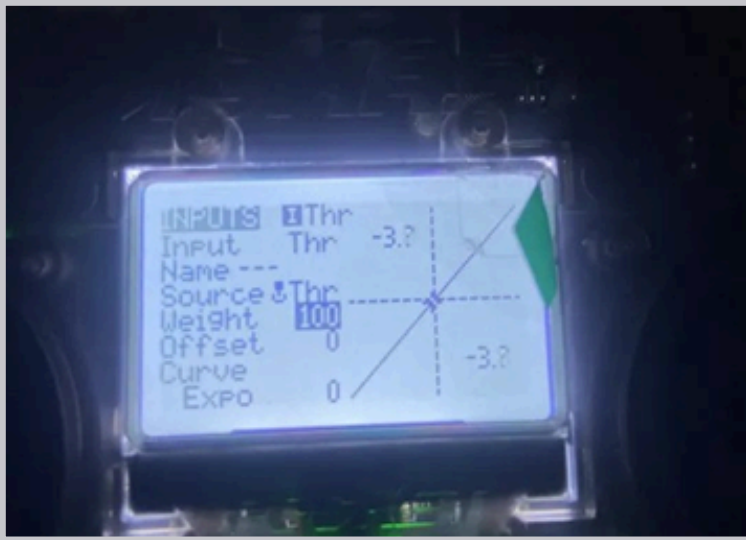


**CHOOSING PROPS-
OUT PROPELLER
CONFIGURATION
BASED ON LEADING
EDGE SO THAT
COLLISIONS DON'T
PULL OBJECTS IN**

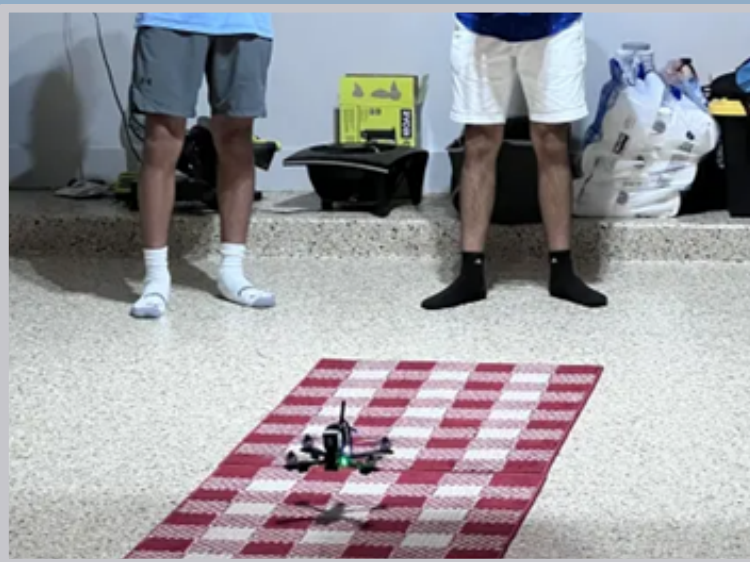


**CALIBRATING
POWER OF
INDIVIDUAL
MOTORS TO
ACCOUNT FOR
STICK SHIFT**

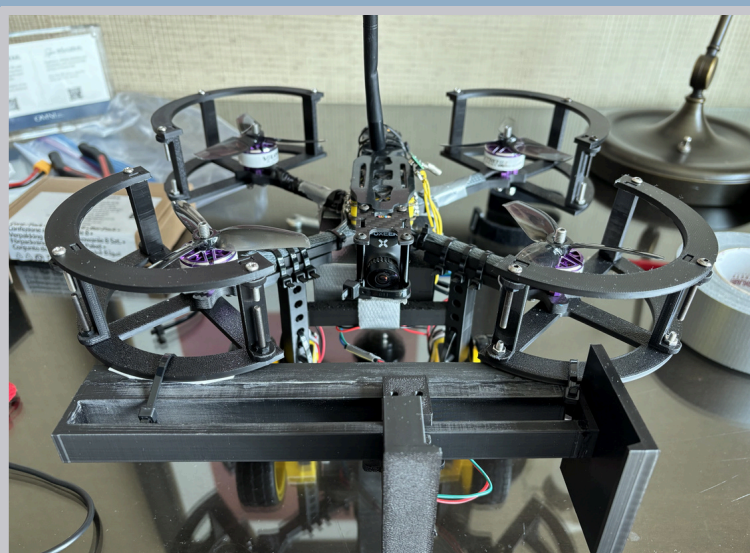
Photo Log (C)



**CALIBRATING
MOTOR THRUST TO
DECREASE
ACCELERATION AND
DECELERATION
SENSITIVITY**

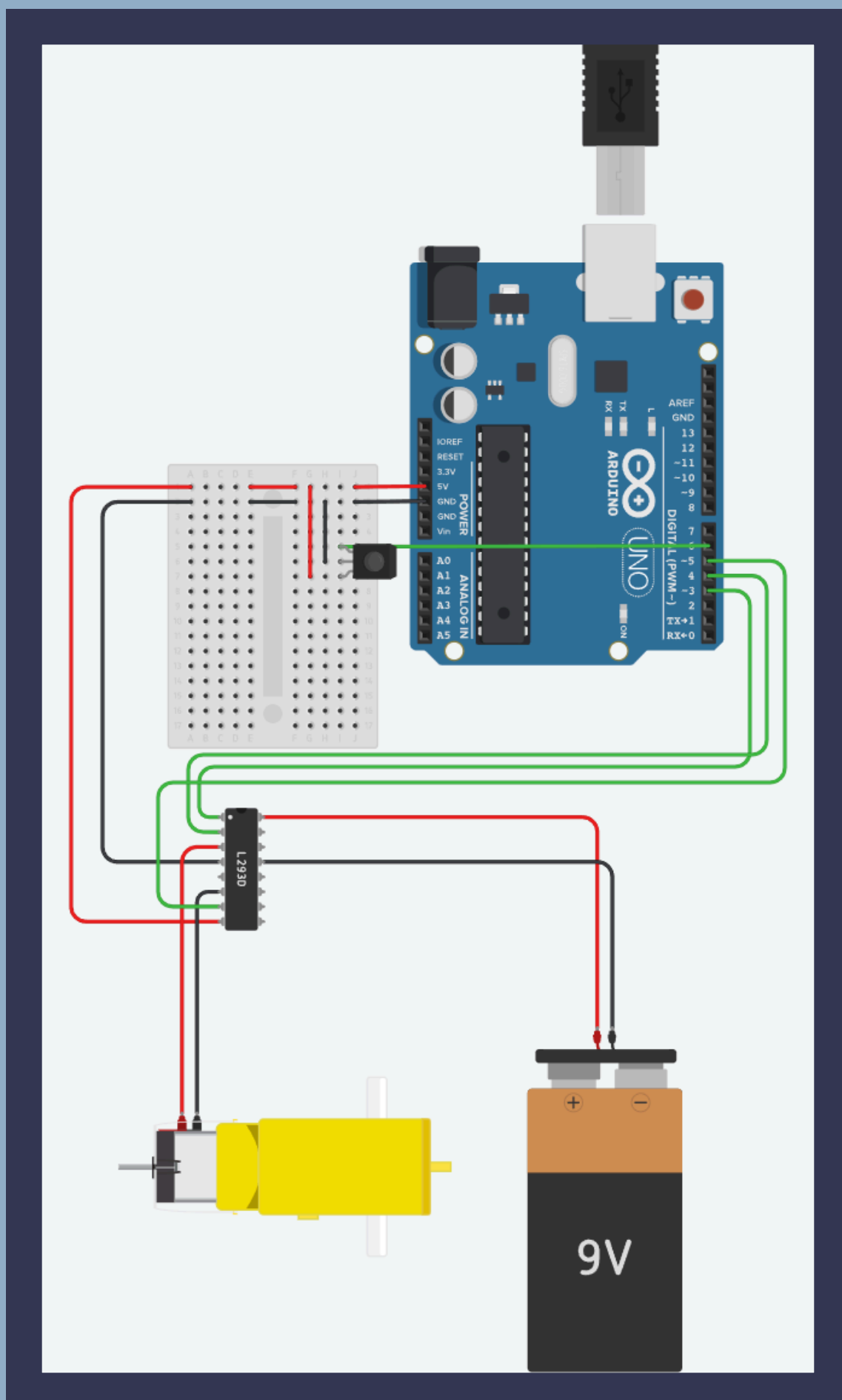


**PRACTICING TAKE-
OFF, LANDING, AND
OTHER FLYING
SKILLS IN HORIZON
MODE TO GET MORE
MANEUVERABILITY**



**ATTACHMENT OF 3D
PRINTED STILTS &
CLAW: FINAL
DRONE BUILD**

Wiring Schematic — Claw (D)



Programming Software (E)

THE SOFTWARE USED FOR FLIGHT FUNCTIONALITY AND STABILIZATION WAS BETAFLIGHT AND ESC-CONFIGURATOR.COM WHICH ARE BOTH OPEN SOURCED AND WERE USED TO CONFIGURE THE DRONE'S FLIGHT CONTROLLER, ESC, AND OTHER FUNCTIONS.

ESC-CONFIGURATOR.COM WAS INITIALLY USED IN ORDER TO FINE-TUNE THE ELECTRONIC SPEED CONTROLLER FOR THE MULTIROTOR DRONE. THIS SOFTWARE IS COMPATIBLE WITH VARIOUS DIFFERENT FIRMWARE OF WHICH "BLUEJAY JH50" IS COMPATIBLE WITH THE ESC USED - SPEEDYBEE F405 V3. IN THIS SOFTWARE, PARAMETERS THAT CAN BE ADJUSTED FOR THE MOTORS INCLUDE MOTOR TIMING, MOTOR DIRECTION, STARTUP POWER, BRAKE FORCE, AND THROTTLE RESPONSE CURVES. CONFIGURING THESE SETTINGS PROPERLY IS CRUCIAL FOR OPTIMIZING THE PERFORMANCE AND EFFICIENCY OF THE MOTORS AND ENSURING SMOOTH OPERATION OF THE AIRCRAFT. WE UTILIZED THIS SOFTWARE BY FIRST PLUGGING A USB-C TO USB-C CABLE BETWEEN THE ESC AND A PORT ON THE LAPTOP ON WHICH THIS SOFTWARE WAS RUN. THEN THE SETTINGS ON THE ESC WERE READ AND "FLASHED" OR UPDATED ONTO ALL OF THE MOTORS. THIS WAS A SORT OF PRE-PROCESSING TO DO BEFORE THE DRONE WAS CONFIGURED WITH BETAFLIGHT.

NEXT, THE BETAFLIGHT SOFTWARE WAS USED TO FURTHER TUNE THE FLIGHT CONTROLLER. FIRST, USING THE AUTO-DETECT FEATURE, THE FLIGHT CONTROLLER (SPEEDYBEE F405 V3) AND ITS VERSION (4.4.3) WAS SELECTED, AND FIRMWARE FOR THE FC WAS LOADED FROM ONLINE. A COMMON PROBLEM THAT OFTEN OCCURS WITH BETAFLIGHT IS THAT THE "PORT IS NOT FOUND" MEANING THAT AN INCORRECT DRIVER HAS BEEN DOWNLOADED. THIS ISSUE PERSISTED WHEN BETAFLIGHT WAS USED SO THE SOFTWARE IMPULSERC DRIVER FIXER BY IMPULSERC WAS USED TO FIND AND ADD THE CORRECT DRIVER SUCH THAT BETAFLIGHT WOULD BE ABLE TO "FLASH" OR UPDATE THE FIRMWARE ON THE FC. THE CHIP WAS PREVIOUSLY ERASED WHILE DOWNLOADING THE FIRMWARE FROM ONLINE.

Programming Software (E)

AFTER THIS, CONFIGURATION BEGINS WITH CHOOSING THE CORRECT PORTS WITH THE IDENTIFIERS "USB VCP", "UART1" (FOR THE VTX OR VIDEO TRANSMITTER), "UART2", AND "UART4". THESE PORTS SERVE AS CHANNELS THROUGH WHICH THE FLIGHT CONTROLLER COMMUNICATES WITH EXTERNAL DEVICES SUCH AS RECEIVERS, GPS MODULES, TELEMETRY TRANSMITTERS, LED STRIPS, AND OTHER ACCESSORIES.

NEXT, THE ACCELEROMETER IS CONFIGURED WITH A SIMPLE BUTTON IN THE SOFTWARE, EMPHASIZING THE SIMPLICITY OF THE PROGRAM FOR USERS. VARIOUS ALGORITHMS DISPLAYED IN A SIMPLE USER INTERFACE ARE ALSO AVAILABLE LIKE THE PID (PROPORTIONAL, INTEGRAL, DERIVATIVE) TUNING CONTROL WHICH HAS MANY PARAMETERS SUCH AS PID GAINS, RATES, EXPO, AND FILTERS WHICH ESSENTIALLY ADJUST THE MOTOR SPEEDS BASED ON THE DIFFERENCE BETWEEN THE DESIRED ORIENTATION AND THE ACTUAL ORIENTATION OF THE AIRCRAFT, MINIMIZING ERRORS AND STABILIZING THE FLIGHT.

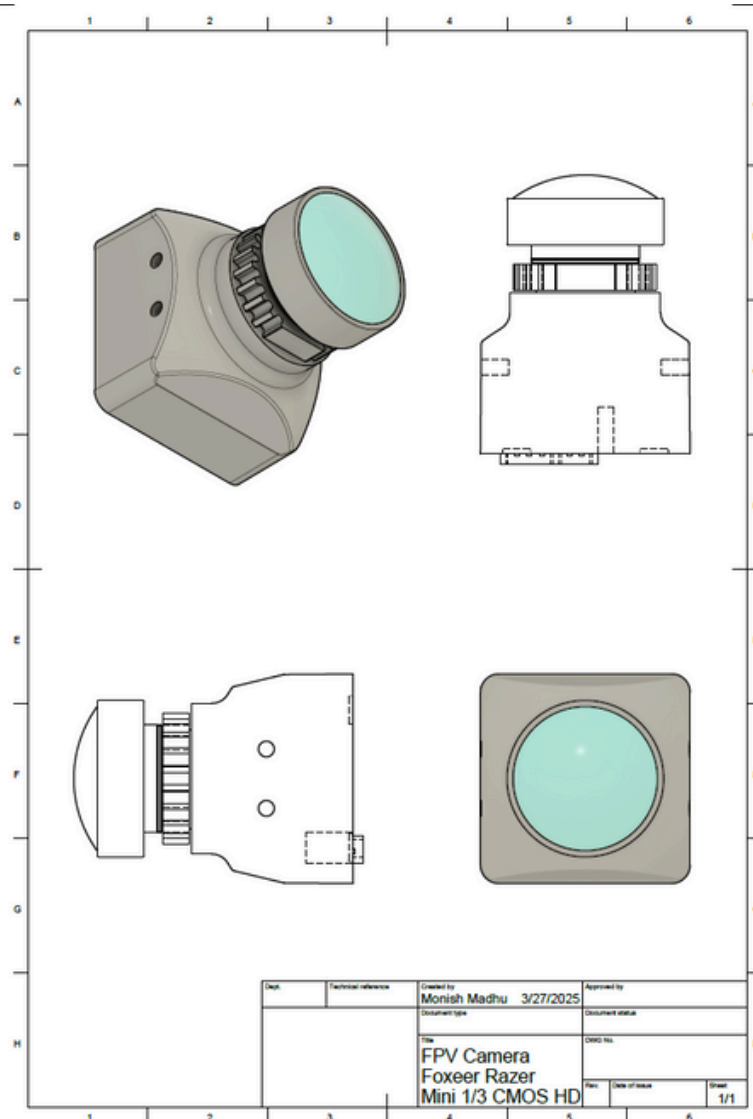
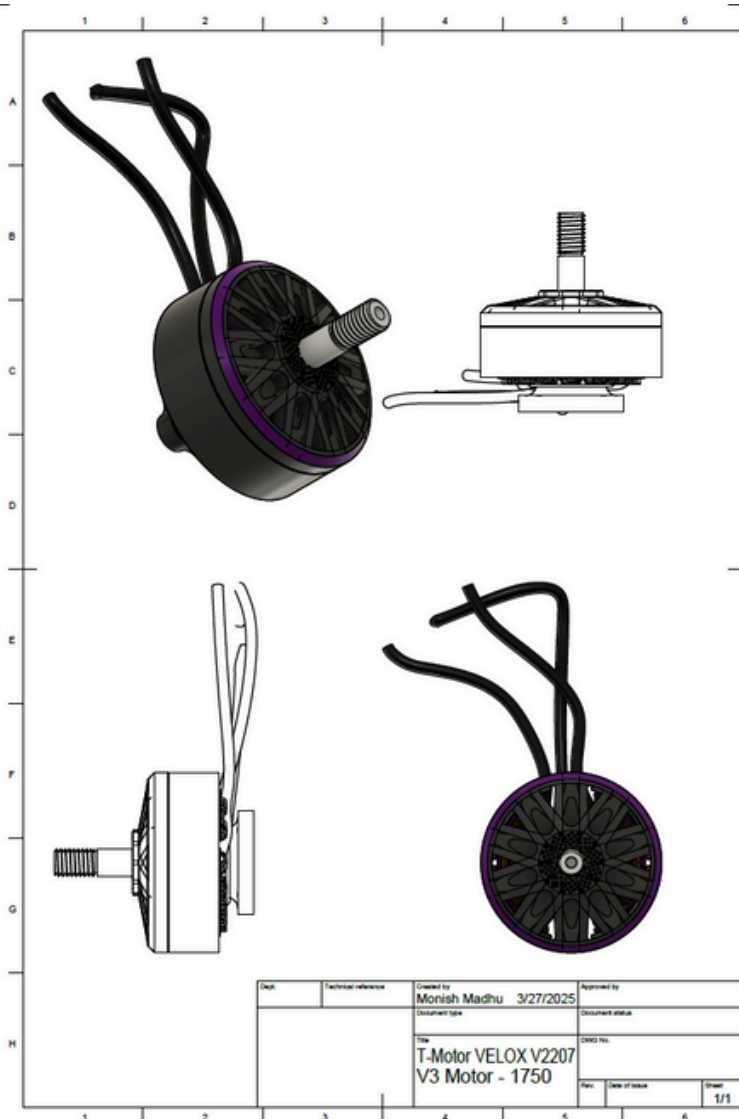
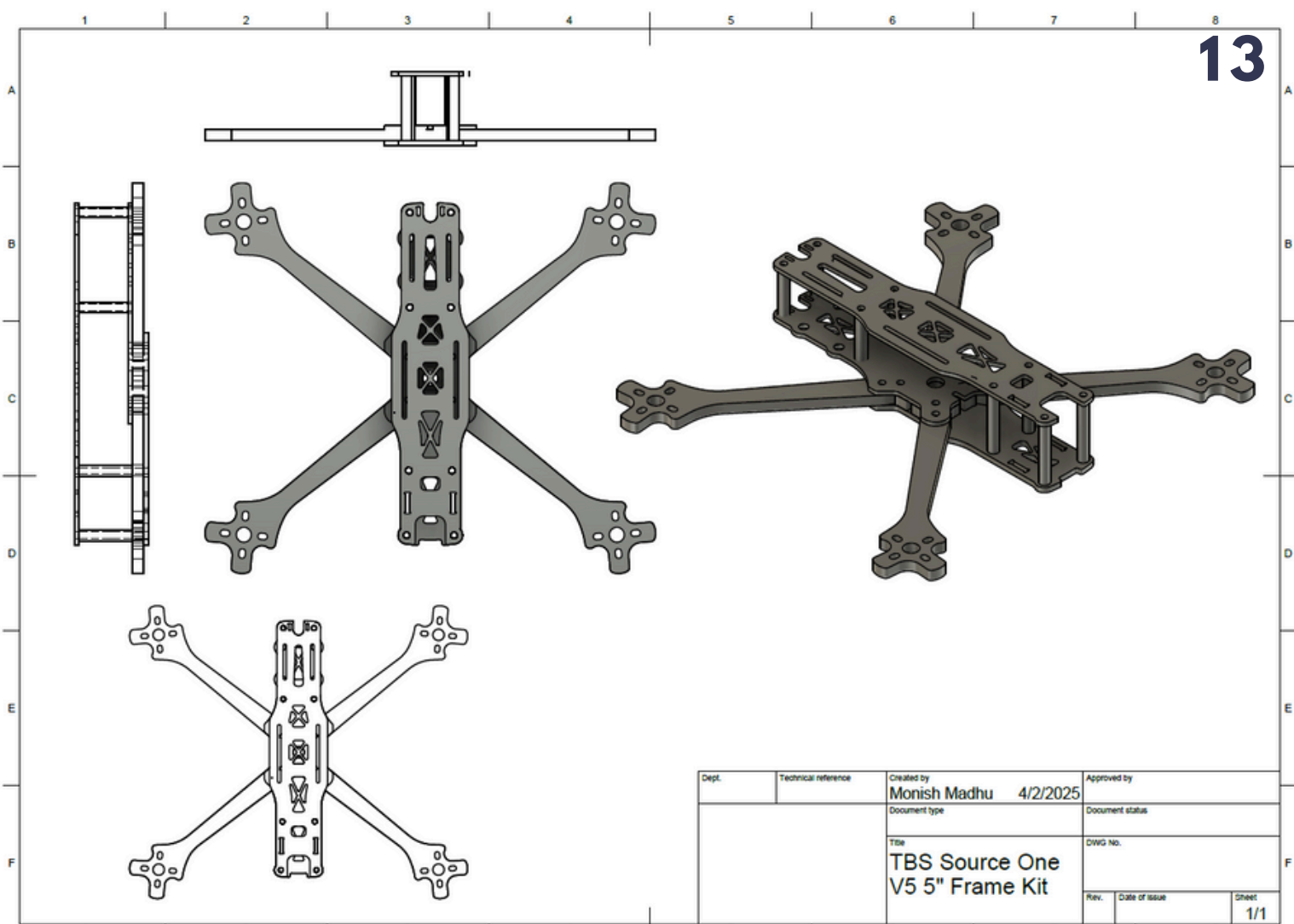
ANOTHER FEATURE USED TO TEST THE DRONE WAS IN THE MOTORS TAB WHERE - USING THE DSHOT300 ESC PROTOCOL FOUND ON THE DRONE'S ESC - VARIOUS FEATURES WERE APPLIED TO THE DRONE LIKE THE ERC_SENSOR AND BIDIRECTIONAL DSHOT. DSHOT (DIGITAL SHOT) IS A DIGITAL COMMUNICATION PROTOCOL DEVELOPED SPECIFICALLY FOR THE COMMUNICATION BETWEEN FLIGHT CONTROLLERS AND ELECTRONIC SPEED CONTROLLERS (ESCS) IN DRONES. UNLIKE TRADITIONAL PROTOCOLS LIKE PWM (PULSE WIDTH MODULATION) OR ONESHOT, WHICH USE ANALOG SIGNALS, DSHOT SENDS DIGITAL SIGNALS, PROVIDING MORE PRECISE AND RELIABLE COMMUNICATION. BIDIRECTIONAL DSHOT IS SIMPLY THE ABILITY TO HAVE TWO-WAY COMMUNICATION BETWEEN THE ESC AND FLIGHT CONTROLLER. THIS TAB ALSO ALLOWED FOR THE TESTING OF THE DRONE MOTORS, THROUGH THE RECONFIGURE MOTOR BUTTON WHICH WOULD ASSIGN A MOTOR TO A LOCATION ON THE DRONE (1 TO 4) AND MANUALLY ARM THE DRONE AND LET US CONFIGURE THE SPIN DIRECTION (CW OR CCW).

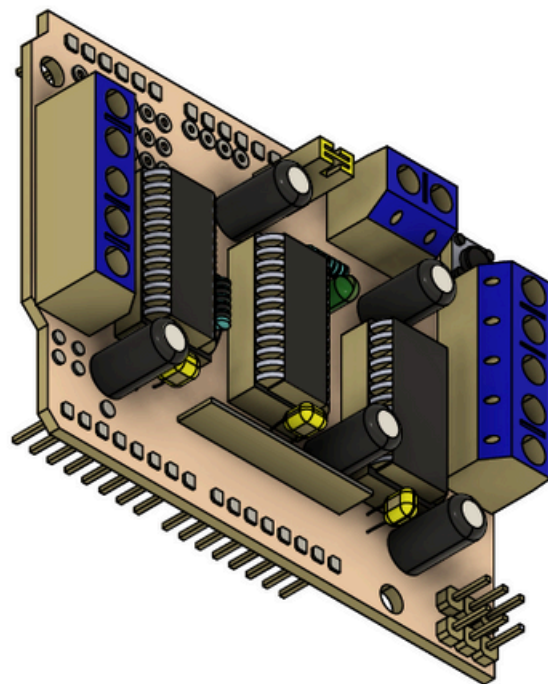
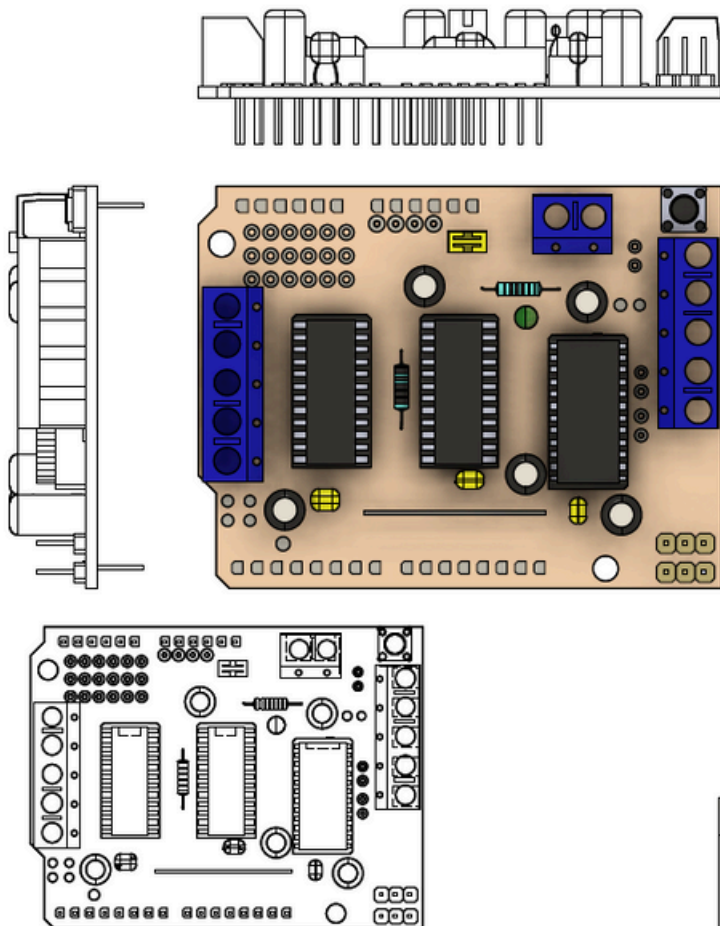
Programming Software (E)

BETAFLIGHT ALSO PROVIDES TELEMETRY SUPPORT, ALLOWING REAL-TIME TRANSMISSION OF FLIGHT DATA TO THE GROUND STATION OR GOGGLES. THIS DATA CAN INCLUDE BATTERY VOLTAGE, CURRENT CONSUMPTION, GPS COORDINATES, ALTITUDE, AND OTHER VITAL PARAMETERS. BETAFLIGHT ALSO SUPPORTS ON-SCREEN DISPLAY (OSD) FEATURES, OVERLAYING FLIGHT INFORMATION DIRECTLY ONTO THE PILOT'S VIDEO FEED FOR ENHANCED SITUATIONAL AWARENESS. WE WERE ABLE TO ADD VOLTAGE/BATTERY WARNINGS AND ALTITUDE AS STATISTICS TO SHOW IN THE BETAFPV DRONE GOGGLE SCREEN.

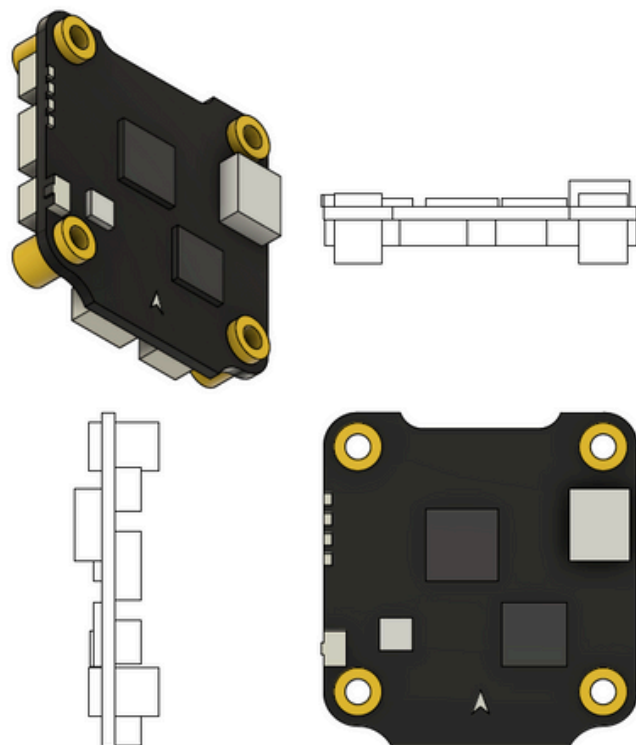
BETAFLIGHT ALLOWS FOR THE ACTUAL MAPPING OF THE CONTROLLER JOYSTICK AND BUTTONS TO MOVEMENTS ON THE DRONE. TO DO THIS, THE RECEIVER TAB IS USED ALONG WITH EXPRESSLRS SOFTWARE (OPENED BY SIGNING INTO THE WIFI EXPRESSLRS WITH PASSKEY EXPRESSION). HERE, A BINDING PHRASE CONNECTS THE EXPRESSLRS TX MODULE THAT IS INTERNAL OR AN EXTERNAL ATTACHMENT TO THE RADIO, AND A EXPRESSLRS RX RECEIVER FOUND ON THE DRONE. AFTER THIS, THE RECEIVER TAB SHOWS VALUES SUCH AS THROTTLE AND MOVEMENT WHILE TESTING STICK MOVEMENT. FROM THERE, THE MODES TAB ALLOWS TO MAP RADIO CONTROLLER BUTTONS TO FUNCTIONS SUCH AS ARMING THE DRONE (TURNING ON THE MOTORS) AND TURNING ANGLE MODE ON OR OFF WHICH RESTRICTS DRONE MOVEMENT.

LASTLY, THE CLAW MECHANISM FUNCTIONS USING SEVERAL ELECTRONIC COMPONENTS, INCLUDING AN ARDUINO MICROCONTROLLER, AN L298N MOTOR DRIVER, A 9V BATTERY, AN IR RECEIVER, AN IR REMOTE, AND A DC MOTOR. THE BATTERY SUPPLIES POWER TO THE SYSTEM, WITH CONNECTIONS ROUTED THROUGH A MINI BREADBOARD. WHEN THE IR REMOTE SENDS A SIGNAL, THE IR RECEIVER DETECTS IT AND RELAYS THE INPUT TO THE ARDUINO. THE ARDUINO THEN PROCESSES THE COMMAND AND INSTRUCTS THE MOTOR DRIVER TO CONTROL THE DC MOTOR'S SPEED AND DURATION. THE MOTOR, DIRECTLY ATTACHED TO THE 3D-PRINTED CLAW, MOVES IT USING A GEAR SYSTEM.



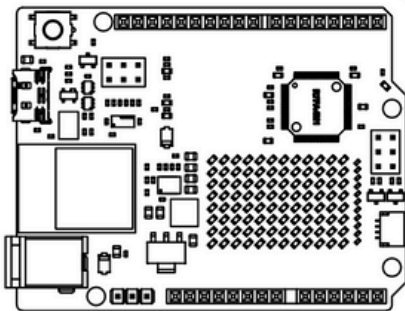
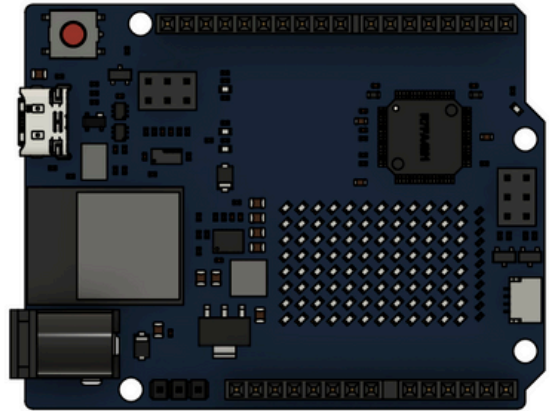
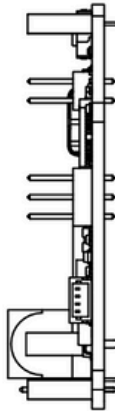
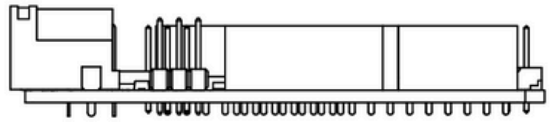
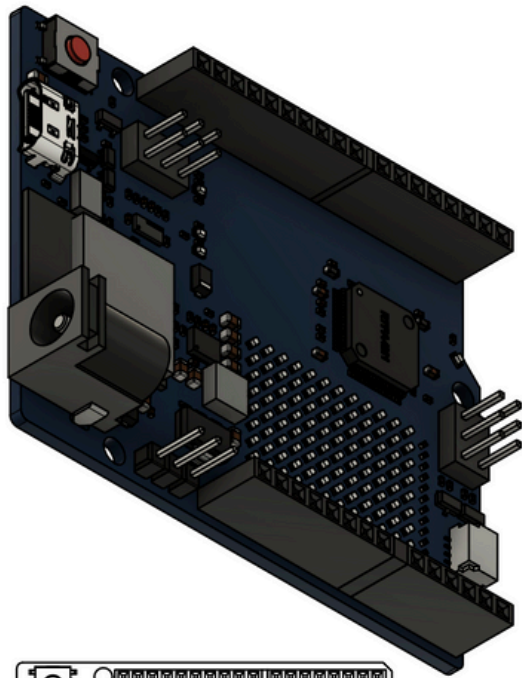


Dept.	Technical reference	Created by Monish Madhu 3/27/2025	Approved by
		Document type	Document status
		Title L293D Motor Driver Shield	DWG No.
		Rev.	Date of issue
			Sheet 1/1

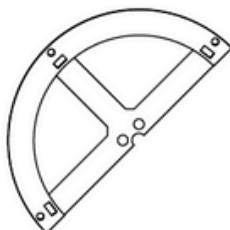
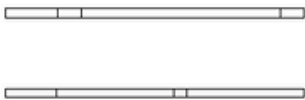


Dept.	Technical reference	Created by Monish Madhu 3/27/2025	Approved by
		Document type	Document status
		Title HappyModel ELRS Nano 2.4 GHz EPI RX (Receiver)	DWG No.
		Rev.	Date of issue
			Sheet 1/1

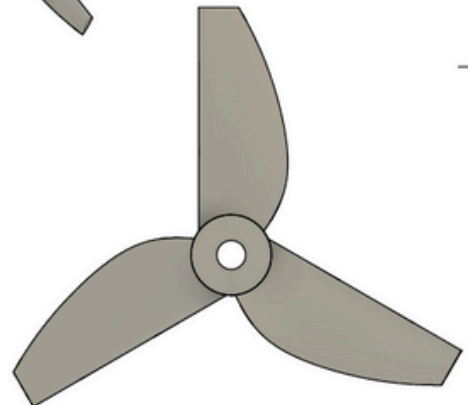
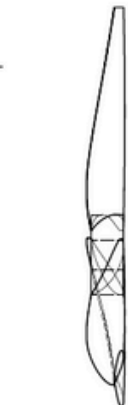
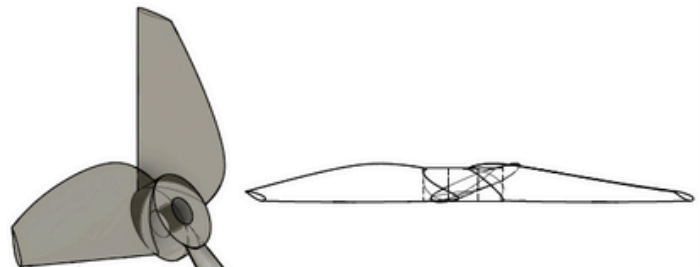
Dept.	Technical reference	Created by Monish Madhu 3/27/2025	Approved by
		Document type	Document status
		Title SpeedyBee F405 Stack - V3	DWG No.
		Rev.	Date of issue
			Sheet 1/1



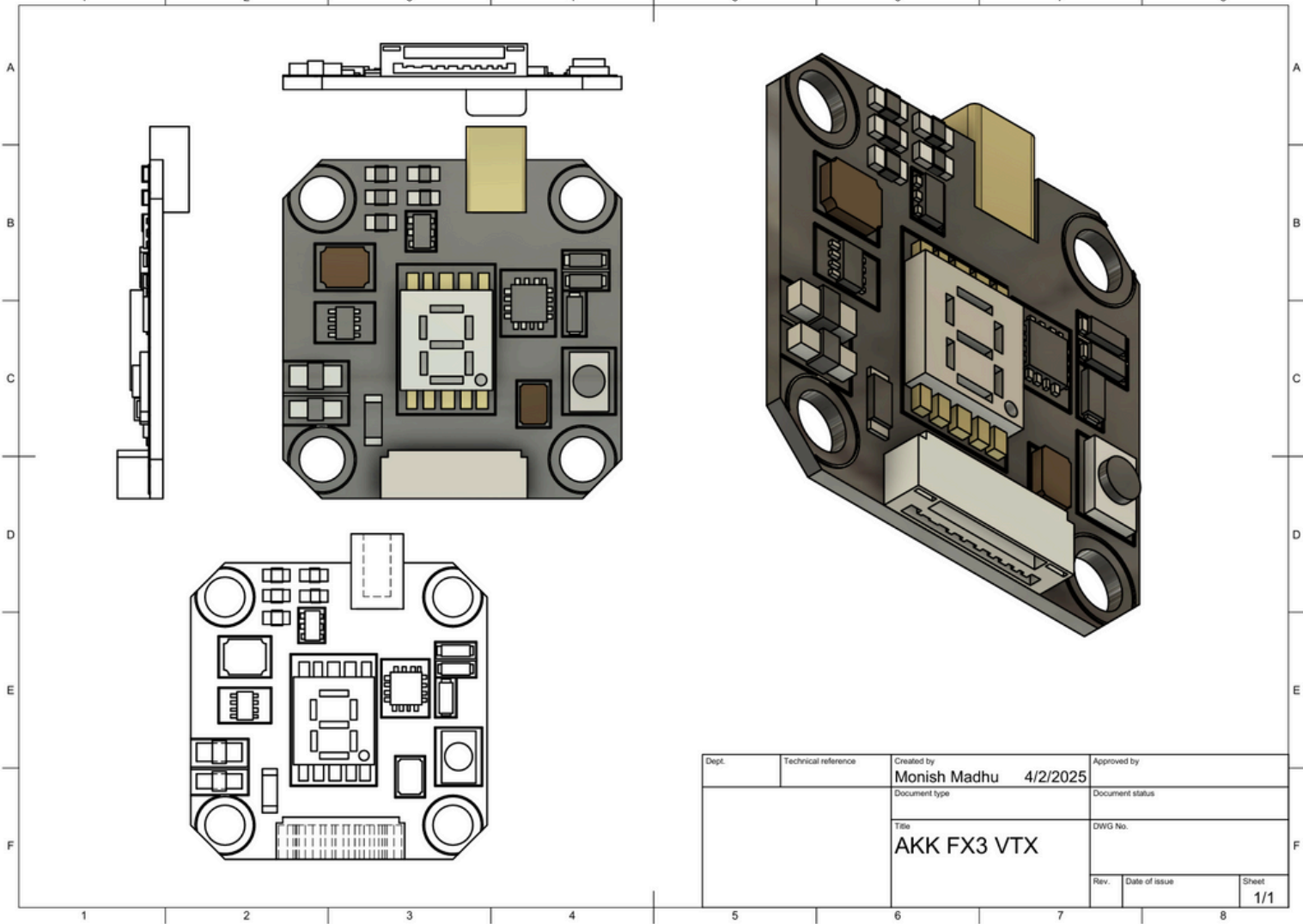
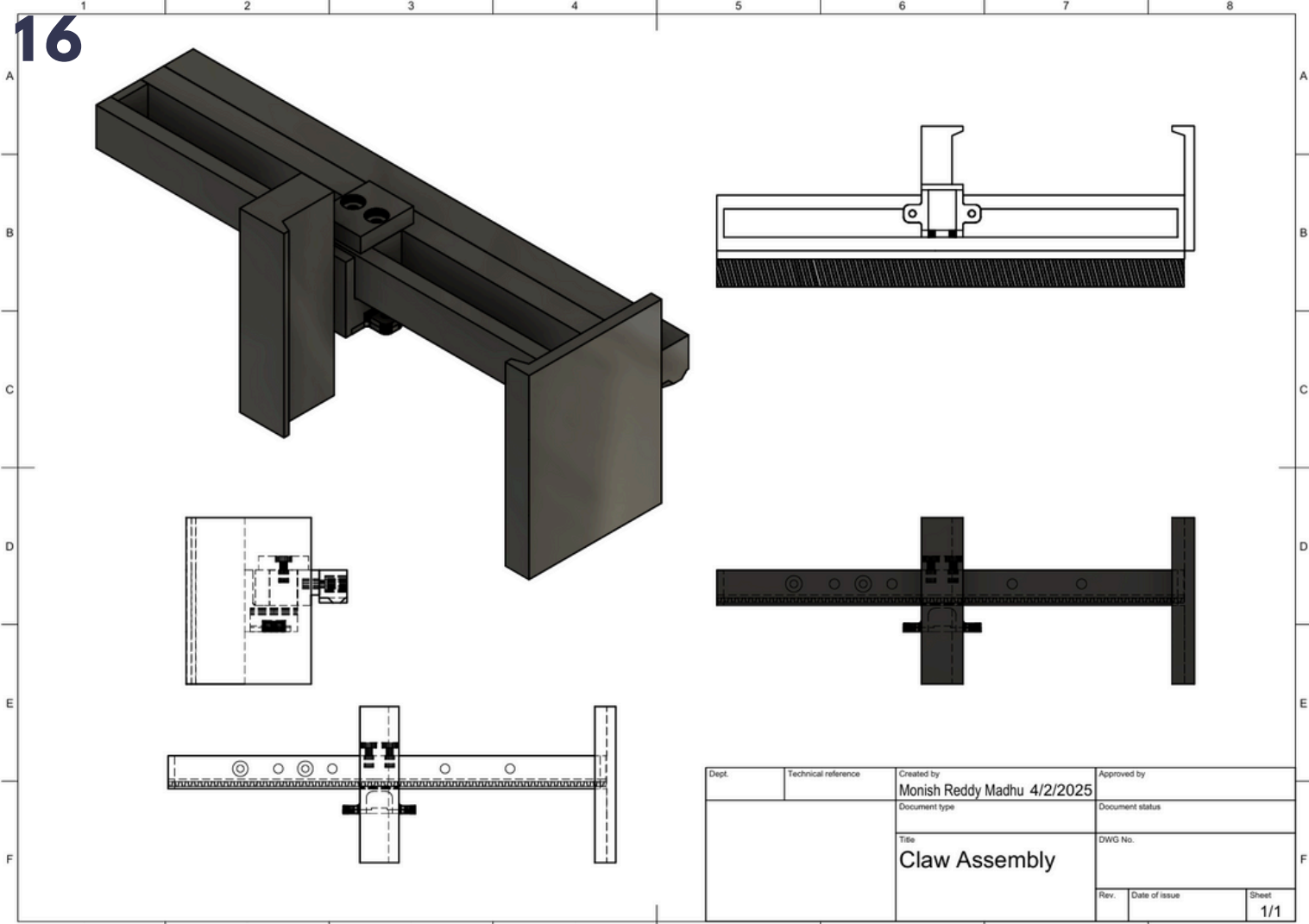
Dept.	Technical reference	Created by Monish Madhu 3/27/2025	Approved by
		Document type	Document status
		Title Arduino UNO R4 WiFi	DWG No.
		Rev.	Date of issue
			Sheet 1/1



Dept.	Technical reference	Created by Monish Madhu 4/2/2025	Approved by
		Document type	Document status
		Title Propeller Guards	DWG No.
		Rev.	Date of issue
			Sheet 1/1



Dept.	Technical reference	Created by Monish Madhu 3/27/2025	Approved by
		Document type	Document status
		Title INTGAMX GEMFAN 4032 4" Propellers	DWG No.
		Rev.	Date of issue
			Sheet 1/1



Bill of Materials (G)

Item #	Part Name	Specifications	Quantity	Cost	Description
1	Drone Frame	TBS Source One V5 5" Frame Kit	1	\$29.99	drone frame made of carbon fiber
2	Flight Controller	SpeedyBee F405 Stack - F405 V3 FC + 50A 3-6s BLHeli_S 4-in-1 ESC - 30x30	1	\$71.99	drone motherboard
3	Motor	T-Motor VELOX V2207 V3 Motor - 1750KV	4	\$14.99	drone's four motors attached to each arm.
4	FPV Camera	FPV Camera Foxeer Razer Mini 1/3 CMOS HD 5MP 2.1mm Lens 1200TVL 4:3 PAL/NTSC Switchable Support 4.5V-25V Input Voltage for FPV Quadcopter Racing Drone	1	\$29.99	FPV HD camera attached to the front of the drone for vision
5	Radio Transmitter	Happymodel ExpressLRS Nano 2.4GHz EP1 RX	1	\$16.99	Receiver to receive inputs from external radio
6	Video transmitter	AKK FX3 5.8Ghz 25mW/200mW/400mW/600mW Switchable FPV VTX with MMCX Integrated OSD FC Configuring via Betaflight Flight Control Board	1	\$18.99	Transmits video data to the flight goggles
7	Mesh wire cover	RDQ 3/8" x 2ft. Braided Mesh Wire Wrap for ESC and Motor Wires	2	\$1.99	wire sleeve to keep wires in place and tidy
8	Battery	Tattu R-Line Version 5.0 22.2V 6S 1200mAh 150C LiPo Battery - XT60	1	\$32.99	power source for drone build
9	Propellers	Azure Power Johnny Freestyle 4.8x3.8x3 POPO Compatible Tri-Blade 5" Prop 4 Pack	1	\$3.29	propellers to keep the drone in the air and flying
10	Claw	3D-Printed Claw Mechanism with Motor	1	\$0.00	Claw to pick up objects

Bill of Materials (G)

Item #	Part Name	Specifications	Quantity	Cost	Description
11	Adjustable Stilts	3D-Printed Landing Stilts & Propeller Guards	4 (each)	\$0.00	allow drone to land without damaging claw mechanism
12	Radio	RadioMaster RadioMaster Pocket 2.4GHz 16CH ELRS/CC2500 Hall Gimbals EdgeTX System Radio Controller Built-in LED Lights Transmitter for FPV	1	\$64.99	external radio to control the drone
13	FPV Monitor	FPV Monitor 5.8 Ghz 4.3" display	1	\$63.99	monitor to connect to the VTX of the drone
14	Battery	LCD Universal 18650 Battery Charger for 3.7V Li-ion Rechargeable Batteries	1	\$11.99	Charger for 18650 3.7V lithium-ion batteries
15	Battery	18650 Rechargeable Battery 3.7V Flat top Lithium Rechargeable Batteries	1	\$15.99	rechargeable lithium-ion cells to power the Radiomaster radio
16	Battery Charger	HTRC LiPo Battery Charger RC Balance Fast Charger Discharger 150W 10A AC/DC	1	\$56.99	charger for the large lithium-ion 22.2V battery
17	Arduino Uno R4	L298N Motor Drive Controller Board Module Dual H Bridge DC Stepper For Arduino	1	\$0.00	programmed to control claw using IR signal
18	L293D Motor Driver	L293D Motor Driver	1	\$16.99	sourced from Arduino kit, used to control DC motor
19	Adapter	2Pair No Wires XT30 to XT60 Plug Female Male Adapter Wireless Connector for RC FPV Drone Car Lipo NiMH Battery Charger ESC	1	\$6.99	adapter to connect battery to the charger
20	Claw Controller	IR Remote	1	\$0.00	sourced from Arduino kit, used to control claw

Rules and Regulations (H)

NASHVILLE, TENNESSEE (NATIONAL CONFERENCE LOCATION)

THE FEDERAL AVIATION ADMINISTRATION (FAA) HAS SET NATIONWIDE REGULATIONS TO ENSURE THE SAFE OPERATION OF DRONES, INCLUDING THOSE FLYING WITHIN NASHVILLE, TENNESSEE. THESE REGULATIONS APPLY TO BOTH RECREATIONAL AND COMMERCIAL DRONE PILOTS AND ARE INTENDED TO PROTECT PUBLIC SAFETY, PROPERTY, AND AIRSPACE INTEGRITY. ADDITIONALLY, TENNESSEE STATE LAWS AND LOCAL ORDINANCES IMPOSE FURTHER RESTRICTIONS TO ADDRESS PRIVACY CONCERNS, INFRASTRUCTURE SECURITY, AND ENVIRONMENTAL PRESERVATION.

FEDERAL REGULATIONS SET FORTH BY THE FAA APPLY UNIFORMLY ACROSS ALL STATES, INCLUDING TENNESSEE. ANY DRONE WEIGHING MORE THAN 250 GRAMS (0.55 LBS) MUST BE REGISTERED WITH THE FAA BEFORE OPERATION, ENSURING ACCOUNTABILITY AND OWNERSHIP TRACKING. RECREATIONAL DRONE PILOTS ARE ALSO REQUIRED TO PASS THE RECREATIONAL UAS SAFETY TEST (TRUST), DEMONSTRATING THEIR AWARENESS OF SAFETY PROTOCOLS. DRONE ALTITUDE IS LIMITED TO 400 FEET ABOVE GROUND LEVEL TO PREVENT CONFLICTS WITH MANNED AIRCRAFT, AND OPERATORS MUST MAINTAIN A DIRECT VISUAL LINE OF SIGHT WITH THEIR DRONES AT ALL TIMES. ADDITIONALLY, DRONES CANNOT BE FLOWN DIRECTLY OVER PEOPLE UNLESS THEY MEET SPECIFIC FAA SAFETY CRITERIA. FLIGHTS NEAR AIRPORTS, INCLUDING NASHVILLE INTERNATIONAL AIRPORT (BNA), REQUIRE PRIOR AUTHORIZATION THROUGH THE FAA'S LAANC (LOW ALTITUDE AUTHORIZATION AND NOTIFICATION CAPABILITY) SYSTEM. DRONES MAY ONLY OPERATE AT NIGHT IF EQUIPPED WITH ANTI-COLLISION LIGHTING VISIBLE FROM AT LEAST THREE STATUTE MILES.

TENNESSEE STATE LAWS BUILD UPON FAA REGULATIONS BY FURTHER RESTRICTING DRONE OPERATIONS IN SENSITIVE AREAS. PRIVACY CONCERNS ARE ADDRESSED BY PROHIBITING THE USE OF DRONES TO RECORD OR PHOTOGRAPH PRIVATE PROPERTY WITHOUT THE OWNER'S CONSENT. ADDITIONALLY, DRONES CANNOT BE FLOWN WITHIN 250 FEET OF CRITICAL INFRASTRUCTURE, INCLUDING POWER PLANTS, WATER TREATMENT FACILITIES, AND TELECOMMUNICATION TOWERS, WITHOUT PRIOR PERMISSION.

Rules and Regulations (H)

WILDLIFE PROTECTION LAWS, AS STATED IN TENNESSEE ADMINISTRATIVE CODE 1660-01-15, PROHIBIT DRONES FROM HARASSING, DISTURBING, OR CAPTURING WILDLIFE. USING DRONES FOR HUNTING, SCOUTING, OR DISTURBING NESTING ANIMALS IS STRICTLY FORBIDDEN.

LOCAL REGULATIONS IN NASHVILLE FURTHER REFINE DRONE USAGE POLICIES TO ADDRESS PUBLIC SAFETY AND ENVIRONMENTAL CONCERNS. THE METRO NASHVILLE PARKS DEPARTMENT PROHIBITS DRONE FLIGHTS IN ALL CITY PARKS EXCEPT FOR DESIGNATED AREAS SUCH AS WARNER PARK, PEELER PARK, AND CANE RIDGE PARK. PILOTS MUST OBTAIN A PERMIT BEFORE FLYING IN THESE DESIGNATED AREAS. SPECIAL EVENT RESTRICTIONS ALSO APPLY, PREVENTING DRONES FROM FLYING OVER OR NEAR LARGE PUBLIC GATHERINGS, SUCH AS CONCERTS, FESTIVALS, OR SPORTING EVENTS, WITHOUT EXPLICIT APPROVAL FROM EVENT ORGANIZERS. MANY AREAS MANAGED BY THE U.S. ARMY CORPS OF ENGINEERS, INCLUDING LAKES AND RESERVOIRS, PROHIBIT DRONE FLIGHTS UNLESS PRIOR AUTHORIZATION IS OBTAINED. RESTRICTED AIRSPACE ZONES INCLUDE GOVERNMENT AND LAW ENFORCEMENT FACILITIES SUCH AS POLICE STATIONS, COURTHOUSES, MILITARY INSTALLATIONS, AND CORRECTIONAL FACILITIES. MAJOR SPORTS AND ENTERTAINMENT VENUES, INCLUDING BRIDGESTONE ARENA AND NISSAN STADIUM, ALSO ENFORCE STRICT NO-FLY ZONES TO ENHANCE SECURITY.

VIOLATIONS OF DRONE REGULATIONS IN NASHVILLE CAN RESULT IN SIGNIFICANT PENALTIES. THE FAA IMPOSES FINES OF UP TO \$27,500 FOR CIVIL VIOLATIONS AND UP TO \$250,000 FOR CRIMINAL OFFENSES. AT THE STATE LEVEL, UNAUTHORIZED SURVEILLANCE USING DRONES CAN RESULT IN A CLASS C MISDEMEANOR CHARGE, LEADING TO FINES AND POTENTIAL JAIL TIME. LOCAL ORDINANCE VIOLATIONS, SUCH AS OPERATING A DRONE IN RESTRICTED PARKS OR OVER LARGE GATHERINGS, CAN RESULT IN CITATIONS, FINES, OR EVEN THE CONFISCATION OF THE DRONE.

Resources (I)

INTOFPV FORUM USED TO HELP SOLVE FAULTY RECEIVER PROBLEM:

[HTTPS://INTOFPV.COM/T-EXPRESS-LRS-CONNECTS-BUT-NO-STICK-MOVEMENT](https://intofpv.com/t-express-lrs-connects-but-no-stick-movement)

VIDEO WALKTHROUGH OF PARTS, CONFIGURATION, AND BUILD PROCESS:

[HTTPS://YOUTU.BE/ZABLSTKXKIM?SI=P58LZEGW7D52TTI5](https://youtu.be/ZABLSTKXKIM?SI=P58LZEGW7D52TTI5)

BETAFLIGHT ERROR DEBUGGING:

[HTTPS://YOUTU.BE/MY6RARSEUYG?SI=9IUJFMFWWH2EYZNN](https://youtu.be/My6RARSEUYG?SI=9IUJFMFWWH2EYZNN)

DRONE FRAME BUILD GUIDE:

[HTTPS://OSCARLIANG.COM/HOW-TO-BUILD-FPV-DRONE/](https://oscarliang.com/how-to-build-fpv-drone/)

3D MODELED CLAW/GRIPPER DESIGN INSPIRATION:

[HTTPS://YOUTU.BE/PDHVFVL-GC4?SI=XWSTK6VMRLQXRQBN](https://youtu.be/PDHVFVL-GC4?SI=XWSTK6VMRLQXRQBN)

UNDERSTANDING FEDERAL DRONE REGULATIONS:

[HTTPS://WWW.FAA.GOV/UAS/RECREATIONAL_FLYERS](https://www.faa.gov/uas/recreational_flyers)

UNDERSTANDING LOCAL/REGIONAL DRONE REGULATIONS (NATIONAL CONFERENCE: NASHVILLE TENNESSEE):

[HTTPS://UAVCOACH.COM/DRONE-LAWS-TENNESSEE/](https://uavcoach.com/drone-laws-tennessee/)

[HTTPS://WWW.FDOT.GOV/AVIATION/UAS.SHTM](https://www.fdot.gov/aviation/uas.shtm)

TRONCAT FPV. (2023, AUGUST 23) BUILD A 6S FREESTYLE FPV DRONE FOR \$200[VIDEO]. YOUTUBE.

[HTTPS://YOUTU.BE/ZABLSTKXKIM?SI=P58LZEGW7D52TTI5](https://youtu.be/ZABLSTKXKIM?SI=P58LZEGW7D52TTI5)

JUAN GONZALEZ GOMEZ. (8 YEARS AGO) 3D PRINTABLE MANUAL ROBOT CLAW[VIDEO]. YOUTUBE.

[HTTPS://YOUTU.BE/PDHVFVL-GC4?SI=XWSTK6VMRLQXRQBN](https://youtu.be/PDHVFVL-GC4?SI=XWSTK6VMRLQXRQBN)